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Department of Electrical, Electronic and Computer Engineering

EAS 410 - Practical 1
Serial Communication and Computer Memory

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1 Changelog

Version	Changes
1.0	First release 2025.
2.0	First release 2026 – updated Figure 4.

2 Introduction

This practical assignment is designed to review the essential skills required to work with FPGAs. Subsequent practical assignments will require students to use these skills to implement computer hardware subsystems, namely random access memory (RAM) and a central processing unit (CPU).

The assignment is divided into two tasks. Task A requires students to implement a universal asynchronous receiver / transmitter (UART) on the FPGA for the purpose of debugging and demonstration. Furthermore, the design of an 8-bit RAM module must be documented (to be implemented in practical 2) for task B.

The prescribed FPGA board is the Terasic DE10-Lite, which is built around the MAX 10 FPGA.

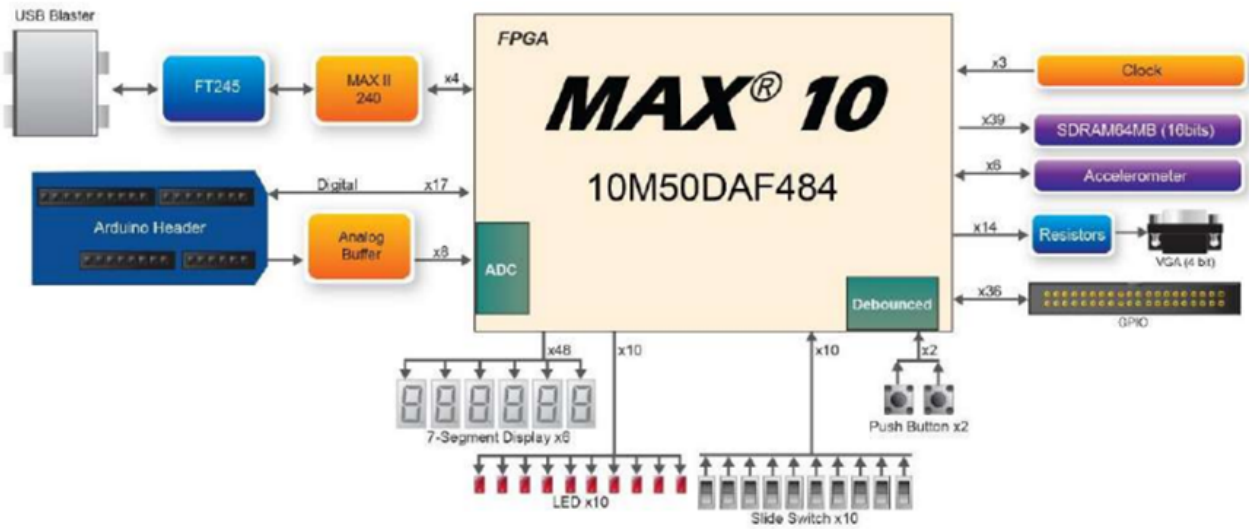

Figure 1: DE10-Lite block diagram [1].

The board's user manual and CD-ROM can be downloaded at [2].

3 Task A

3.1 Overview

A UART must be implemented on the FPGA to complete this task. Its purpose is to facilitate a bidirectional communication link between the FPGA and a PC such that hardware implemented on the FPGA can be controlled for debugging and demonstration purposes. It follows that an application running on the PC, acting as the user interface, must also be developed.

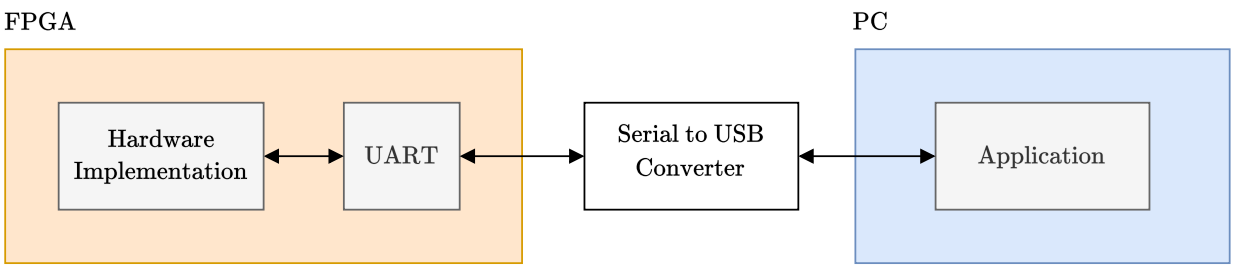

Figure 2: Task A architectural block diagram.

In figure 2, the application running on the PC must generate a sequence of random bytes. It then transmits the bytes to the FPGA which receives the data using the UART. The hardware implementation for this practical is a circular bit shift which is performed on each byte. The shifted bytes are then transmitted back to the PC using the UART and displayed on the PC. Figure 3 shows this procedure.

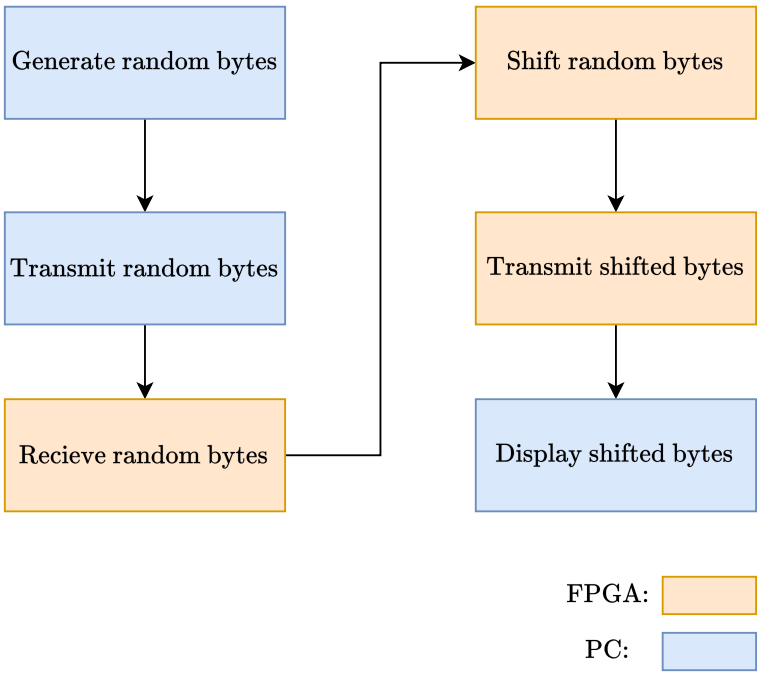

Figure 3: Task A flow diagram.

3.1.1 UART

The UART must be implemented using the following frame/packet specifications:

- 1 start bit

- 8 data bits
- 1 stop bit
- No parity bit
- No flow control
- LSB first

Students will be rewarded for implementations that demonstrate the ability to handle higher baud rates and receive an increased number of bytes.

3.1.2 FPGA Hardware Implementation

The focus for this practical is the UART. Therefore, only a rudimentary hardware function must be implemented - a circular bit shift. The user must be able to control the parameters of the bit shift using the slide switched on the FPGA board.

SW0	**SW1**	**Action**
0	0	One left shift.
0	1	One right shift.
1	0	Two left shifts.
1	1	Two right shifts.

Table 1: Bit shift parameters for each SW0 and SW1 value

For example, if SW0 and SW1 are both 0, and the byte received by the FPGA is 1101 0000, then the resulting byte is 1010 0001.

3.1.3 PC Application

The PC application must be capable of generating a sequence of random bytes, transmitting them to the FPGA and receiving the result of the bit shift. The application must also report when an error has occurred, i.e. when the received byte is incorrect, and the success rate as below.

```
1 0000 0000 - 0000 0000
2 0101 0101 - 1010 1010
3 1111 0000 - 1110 0001
4 1100 1100 - 1001 1000 Error!
                .
                .
                .
47 0001 1111 - 0011 1110
48 0110 1010 - 1101 1000
49 0001 1000 - 0011 0000
50 0100 0011 - 1000 0110

Errors: 1
Transmit Time: 4 ms = 50 x 8 bits / 0.004 s = 100 000 bps
Receive Time: 5 ms = 50 x 8 bits / 0.005 s = 80 000 bps
```

Figure 4: Example of the application's terminal output.

3.2 Detailed Design Information

Task A may be completed using three different approaches. The student's mark is limited if simpler approaches are used, with more complex approaches awarding higher marks.

3.2.1 Approach 1 (Max 100%)

This approach enables students to achieve full marks for task A. To qualify, the UART implementation must meet the following requirements:

- The receiver and transmitter subsystems must be **fully integrated and functional**.
- The system must support a minimum **baud rate of 115200**.
- The implementation must demonstrate the ability to reliably transmit and receive **at least 50 bytes** in a single operation.

This approach follows the flow in figure 5 below. The PC application must generate 50 random bytes, send them to the FPGA, which shifts the bytes according to table 1. The shifted bytes are then transmitted back to the PC and displayed as in figure 4

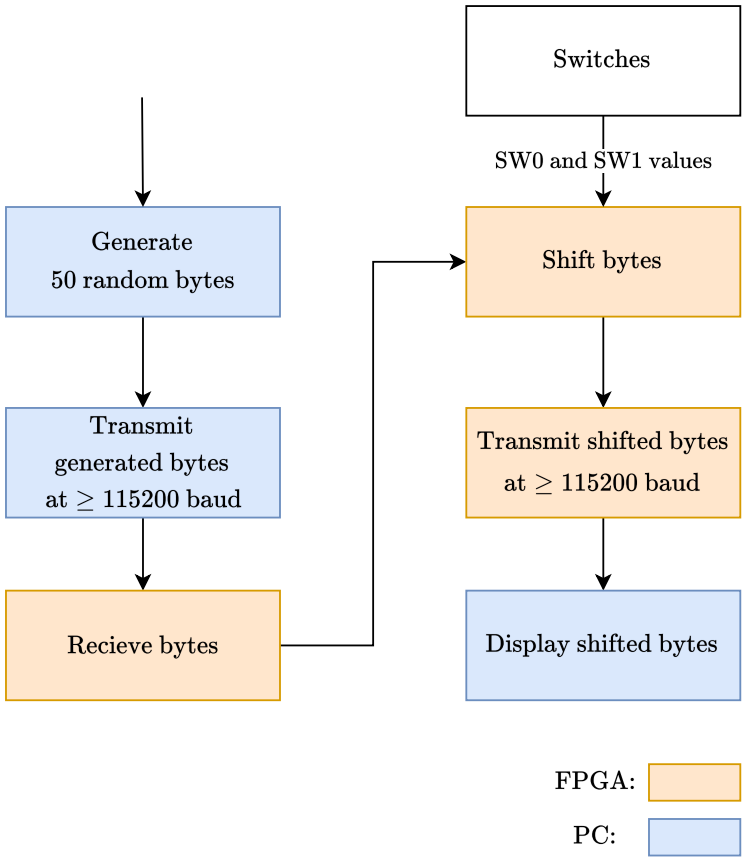

Figure 5: Approach 1 flow diagram.

3.2.2 Approach 2 (Max 75%)

This approach limits the student's mark to 75% for task A. To qualify, the UART implementation must meet the following requirements:

- The receiver and transmitter subsystems must be **fully integrated and functional**.
- The system may use **any baud rate**.
- The implementation must demonstrate the ability to reliably transmit and receive **at least 8 bytes** in a single operation.

For the approach, **the bytes need not be generated**. An 8 byte sequence will be given during the demonstration. The approach follows the flow in figure 6 below. The PC application must be used to send the bytes to the FPGA, which shifts the bytes according to table 1. The shifted bytes are then transmitted back to the PC and displayed as in figure 4.

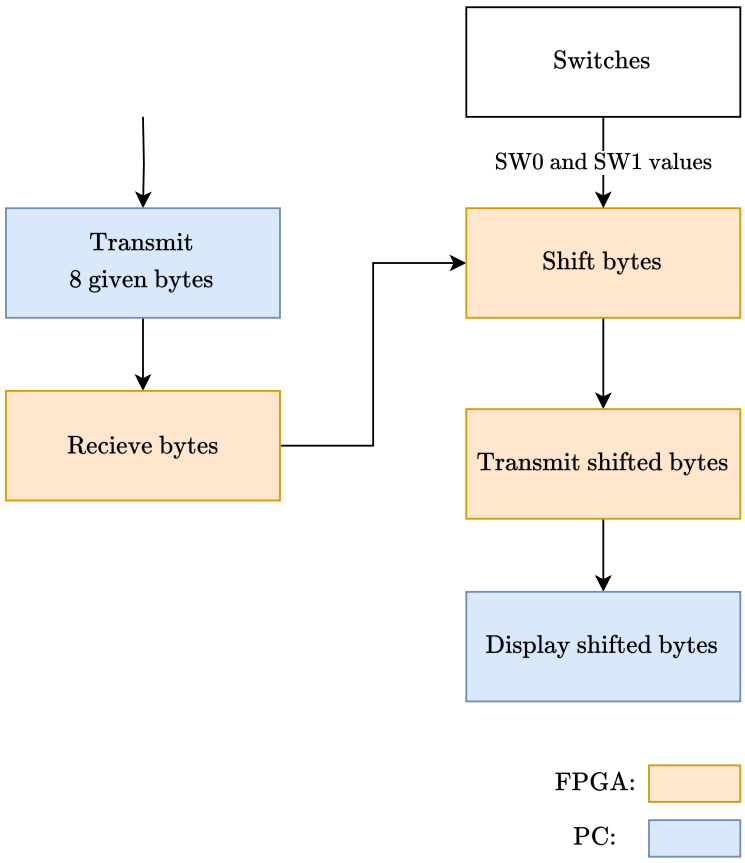

Figure 6: Approach 2 flow diagram.

3.2.3 Approach 3 (Max 50%)

This approach limits the student's mark to 50% for task A. The UART implementation must meet the following requirements:

- The receiver and transmitter subsystems **may be separate** i.e. two different implementations.
- The system may use **any baud rate**.
- The implementation must demonstrate the ability to reliably transmit and receive **one byte** in a single operation.

For this approach, **the bytes need not be generated, nor do they need to be shifted** by the FPGA. A byte will be given during the demonstration. The flow in figure 8 must be followed. The user must be able to **program the given byte using the FPGA switches**.

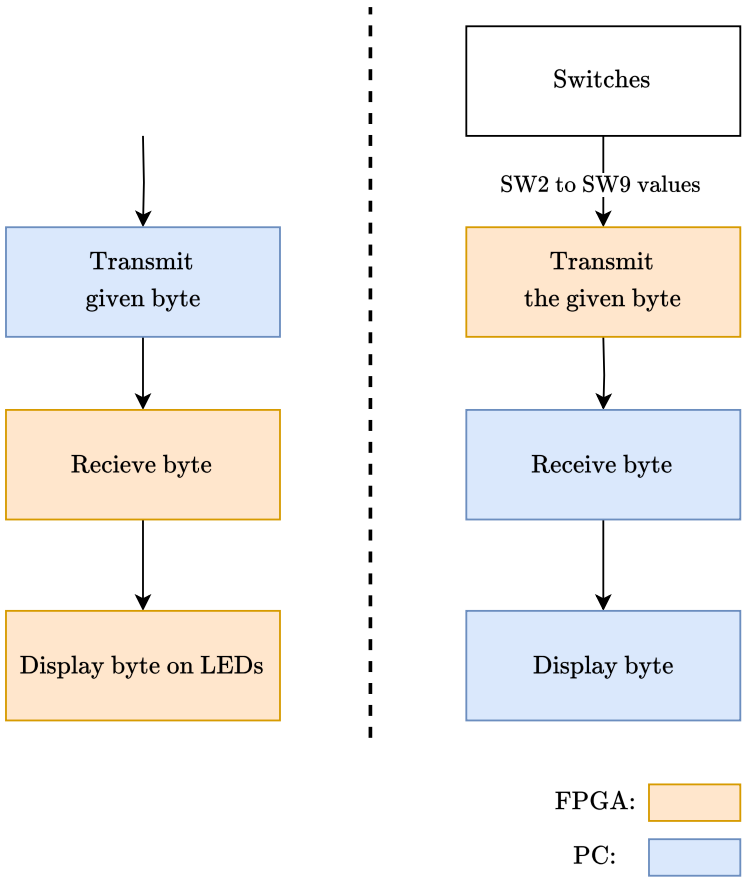

Figure 7: Approach 3 flow diagram.

4 Task B

The design of an 8-bit, 256 byte RAM must be documented to complete this task. The RAM must be designed using an 8x8x1 bit memory chip, i.e. the chip is an 8 by 8 array of bits. The design must allow for full word and half word (nibble) read / writes. This task builds on the example given in chapter 6 of the textbook.

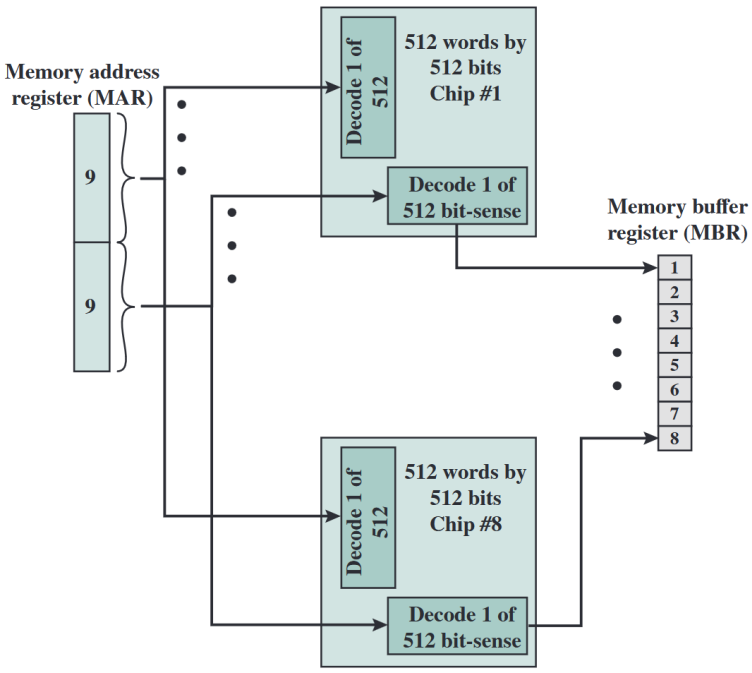

Figure 8: Memory architecture example [3]

5 Summary of Findings

There are no formal reports for EAS 410 practical assignments, however a summary of findings is required for each practical. The required content for practical 1 is as follows:

- Task A
  - VHDL / Verilog snippets showing and explaining all modules created and how they are created.
  - Functional simulations of the UART.
  - Flow diagrams / pseudo code for the PC application.
  - Reasoning for design decisions made.
- Task B
  - The architectural block diagram of the designed RAM
  - Explanations and design decisions made

6 Demonstration

Each group will demonstrate Task A (maximum 10 minutes) and Task B (maximum 5 minutes). Prior to the demonstration, groups will receive a set of eight bytes along with SW0 and SW1 values.

A practical time slot will be assigned to each group and posted the day before the demonstration. Demonstrations will take place on **5 March 2025**, in **Netlab A**. Ensure your hardware is fully prepared before your scheduled time. During the demonstration, be prepared to:

- Present and explain your code.
- Provide lab books (each student must have their own).

Following the hardware demonstration, group members will be asked questions about the practical and their results. All members are expected to understand the entire practical, including designs, calculations, and theoretical concepts, regardless of individual contributions. Responses will contribute to a mark reflecting the group's overall understanding.

Important Note: Every student is responsible for knowing all aspects of the practical, even those they did not directly work on. This includes familiarity with all sections, designs, and theoretical foundations.

7 Code Submission

Groups are required to upload their code to the AMS. The complete code for task A should be uploaded before your demo slot on day of the demonstration to the demonstration assessment on AMS in a zip file with the naming convention **EAS410_P1_*Groupnumber***.

8 Tips

- An Arduino or similar may be used as a usb-to-serial converter.
- Unused FPGA peripherals may be used as part of your UART implementation. For example, using a push button to reset.

9 Additional Information

Please feel free to email me at *up.eas410@gmail.com* if you have any questions. Please also ask questions on Telegram. I will ensure that all the required information pertaining to this practical is uploaded and communicated to you by means of ClickUP.

References

[1] Terasic, *DE10-Lite User Manual*, 2018, v1.6.

[2] "Terasic DE10-Lite Board - Resources." [Online]. Available: https://www.terasic.com.tw/cgi-bin/page/archive.pl?Language=English&CategoryNo=234&No=1021&PartNo=4#contents

[3] W. Stallings, *Computer Organization and Architecture: Designing for Performance*, 11th ed., ser. Global Edition. Pearson Education Limited, 2022.